

Gillian D. Beltz-Mohrmann, Ph.D.

Department of Physics and Program in Statistical & Data Sciences
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Professional Appointments	Assistant Professor - Smith College	2025-
	Department of Physics and Program in Statistical & Data Sciences	
	Postdoctoral Research Fellow - Argonne National Laboratory	2022-2025
	Cosmological Physics and Advanced Computing Group	
	Graduate Research Assistant - Vanderbilt University	2016-2022
	Department of Physics & Astronomy	
Education	Vanderbilt University, Nashville, TN	May 2022
	Ph.D. in Astrophysics	Advisor: Andreas Berlind
	Thesis: <i>Developing an Accurate Probe of the Galaxy-Halo Connection: Baryonic Effects, Small-Scale Galaxy Clustering, and Halo Model Extensions</i>	
	Wellesley College, Wellesley, MA	May 2016
	B.A., <i>cum laude</i>	Advisors: Kim McLeod, James Battat
	Major: Astrophysics; Minor: German	
Honors & Awards	Most Outstanding Student Publication Award	2020
	Vanderbilt Physics & Astronomy Department	
	Graduate Student Poster Competition Winner	2019
	Vanderbilt Data Science Symposium	
	Akunuri V. Ramayya Award for Outstanding Teaching Assistant	2018
	Vanderbilt Physics & Astronomy Department	
	Provost Graduate Fellowship	2016–2021
	Vanderbilt University	
Undergraduate Chambliss Achievement Honorable Mention	2016	
	227th American Astronomical Society Meeting	
Sarah Frances Whiting Medal for Achievement in Astronomy	2014	
	Wellesley College	
Grants	XSEDE Grant	2019, 2020
	Awarded 58.4k total Node Hours (2.8M CPU hours) on Stampede2	
	McMinn Research Grant	2019, 2020
	Vanderbilt Physics & Astronomy Department (\$3,000 total)	
Graduate Summer Research Award	2018	
	Vanderbilt College of Arts and Sciences (\$1,900)	
Teaching	Smith College	
	PHYS 211 Computational Method in the Physical Sciences	Spring 2026
	PHYS 210 Mathematical Methods of Physical Sciences & Engineering	Spring 2026
	SDS 271 Programming for Data Science in Python	Fall 2025
	SDS 291 Multiple Regression	Fall 2025
	Conference for Undergraduate Women in Physics	
	Python workshop	Jan. 2023
	Vanderbilt University	
	Astronomy Lab Instructor	Fall 2016 – Spring 2019
	Wellesley College	
Supplemental Instruction Leader	Fall 2014 – Spring 2016	
Mentoring	Emma Schultz, Smith College	Spring 2026 -
	Topic: Modeling IGM attenuation with Jax	
	Madeline Dean, Smith College	Fall 2025 -

Topic: Modeling galaxies with HODs in Λ CDM and w_0w_a CDM simulations
Zhiwen Ji, Smith College Fall 2025
 Topic: Investigating the concentration-halo mass relation in IllutrisTNG
Harmandeep Gill, University of Toronto, undergraduate Oct. 2024 - May 2025
 Topic: Modeling Lyman Break Galaxies
Ivan Kraskov, University of Toronto, undergraduate Oct. 2024 - Oct. 2025
 Topic: Modeling IGM attenuation in Jax
Emily Martsen, University of Chicago, graduate Sep. 2024 - June 2025
 Topic: Measuring the Two-point Clustering of Galaxy Clusters
Resherle Verna, UT Austin, graduate Summer 2023
 GEM Fellowship Program
 Topic: Forward modeling galaxy SEDs with Jax
Caleigh Dennis, Harpeth Hall High School Sep. 2017 – May 2019
 Topic: Measuring the rotation of galaxy groups in SDSS
 1st place winner at Middle Tennessee Science & Engineering Fair (2018 & 2019)

**1st & 2nd
Author
Publications**

Submitted & Published

Total Citations: 121

8. **Beltz-Mohrmann, G. D.**, Pope, A., et al., 2025, “Illuminating the Physics of Dark Energy with the Discovery Simulations,” The Open Journal of Astrophysics, 8, 74
7. Pearl, A. N., **Beltz-Mohrmann, G. D.**, Hearin, A. P., 2024, “DiffOpt: Parallel optimization of Jax models,” Journal of Open Source Software, 9(104), 7522
6. **Beltz-Mohrmann, G. D.**, Szewciw, A. O., Berlind, A. A., Sinha, M., 2023, “Toward Accurate Modeling of Galaxy Clustering on Small Scales: Halo Model Extensions and Lingering Tension,” The Astrophysical Journal, 948, 100
5. Szewciw, A. O., **Beltz-Mohrmann, G. D.**, Berlind, A. A., Sinha, M., 2021, “Toward Accurate Modeling of Galaxy Clustering on Small Scales: Constraining the Galaxy-Halo Connection with Optimal Statistics,” The Astrophysical Journal, 926, 15
4. **Beltz-Mohrmann, G. D.**, Berlind, A. A., 2021, “The impact of baryonic physics on the abundance, clustering, and concentration of halos,” The Astrophysical Journal, 921, 112
3. **Beltz-Mohrmann, G. D.**, Berlind, A. A., Szewciw, A. O., 2020, “Testing the Accuracy of Halo Occupation Distribution Modelling using Hydrodynamical Simulations,” Monthly Notices of the Royal Astronomical Society, 491, 5771
2. Dale, D. A., **Beltz-Mohrmann, G. D.**, et al., 2016, “Radial Star Formation Histories in Fifteen Nearby Galaxies,” The Astronomical Journal, 151, 4
1. Souza, S. P., **Beltz-Mohrmann, G.**, Sami, M., 2014, “The Light Curve and Period of MT696,” The Journal of the American Association of Variable Star Observers, 42, 154

**Nth Author
Publications**

Submitted & Published

Total Citations: 54

5. Lange, Johannes U. et al., 2025, “Cosmological Constraints from Full-Scale Clustering and Galaxy-Galaxy Lensing with DESI DR1”, arXiv:2512.15962
4. Alarcon, Alex et al., 2025, “DiffstarPop: A generative physical model of galaxy star formation history”, submitted to The Open Journal of Astrophysics
3. OpenUniverse Collaboration et al., 2025, “OpenUniverse2024: A shared, simulated view of the sky for the next generation of cosmological surveys”, Monthly Notices of the Royal Astronomical Society
2. Lange, Johannes U. et al., 2024, “Systematic Effects in Galaxy-Galaxy Lensing with DESI”, The Open Journal of Astrophysics, 7, 57

1. Yuan, Sihan et al., 2024, “Redshift evolution and covariances for joint lensing and clustering studies with DESI Y1”, Monthly Notices of the Royal Astronomical Society, 533, 1

Recent Talks

Five College Astronomy Department Colloquium , UMass Amherst	Feb. 2026
<i>Improving Models of Small-Scale Galaxy Clustering to Constrain Cosmology</i>	
Mock NYC , Center for Computational Astrophysics	Jan. 2026
<i>The DESI Emulator Mock Challenge</i>	
Cosmology from Lyman-Break Galaxies , University of Toronto	May 2025
<i>Modeling the Galaxy-Halo Connection of LBGs</i>	
ELG Mock Challenge , Donostia International Physics Center	Feb. 2025
<i>Lessons Learned from the DESI Emulator Mock Challenge,</i>	
<i>Simulation-based Forward Modeling with Diffsky</i>	
Winter DESI Meeting , Cancun, Mexico	December 2024
<i>Updates on the DESI Emulator Mock Challenge</i>	
Taco Talk , University of Arizona	October 2024
<i>A Differentiable Forward Model of the Galaxy-Halo Connection</i>	
Cosmology Talks Miniworkshop (invited expert)	August 2024
<i>Cosmology Beyond 2pt Statistics</i>	
DHWFEST , University of Utah	July 2024
<i>A New Forward Model of the Galaxy-Halo Connection</i>	
Summer DESI Meeting , Marseille, France	July 2024
<i>DESI Alternative Clustering Methods</i>	
New Strategies for Extracting Cosmology from Galaxy Surveys	July 2024
Sexten Center for Astrophysics	
<i>Simulation-based Forward Modeling of Cross-Survey Cross-Correlations with Diffsky</i>	
Fundamental Physics from Future Spectroscopic Surveys	May 2024
Lawrence Berkeley National Lab	
<i>Making multi-wavelength, multi-redshift predictions for Cross-Survey</i>	
<i>Cosmological Analyses</i>	
Winter DESI Meeting , Hawaii, USA	Dec. 2023
<i>Introducing DESI-Diffsky: A Differentiable Forward Model for Making</i>	
<i>Multi-wavelength, Multi-tracer DESI Mocks</i>	
Building a physical understanding of galaxy evolution with data-driven astronomy , KITP, UC Santa Barbara	Jan. 2023
<i>Toward Accurate Modeling of Galaxy Clustering on Small Scales:</i>	
<i>Halo Model Extensions & Lingering Tension</i>	
CAMELS Workshop , Center for Computational Astrophysics	Dec. 2022
<i>Toward Accurate Modeling of Galaxy Clustering on Small Scales:</i>	
<i>Halo Model Extensions & Lingering Tension</i>	
N-Body Shop Workshop , Center for Computational Astrophysics	June 2022
<i>Accurate Modeling of Galaxy Clustering on Small Scales</i>	
High-Energy and AstroPhysics Seminar , University of Utah	Jan. 2022
<i>Developing an Accurate Probe of the Galaxy-Halo Connection</i>	
KICP Seminar , University of Chicago	Nov. 2021
<i>Developing an Accurate Probe of the Galaxy-Halo Connection</i>	
Galaxies and AGN Journal Club , Johns Hopkins University	July 2021
<i>Impact of baryonic physics on the abundance, clustering, & concentration of halos</i>	
Galaxy Lunch talk , Yale University	March 2021
<i>Can we ignore baryons in halo modeling?</i>	
Galaxy-Halo Connection Across Cosmic Time	Aug. 2020
KITP, UC Santa Barbara	
<i>HMF Discrepancies between Hydrodynamic and DMO Simulations</i>	
Mock Innsbruck (remote participant)	March 2020
<i>Taking Halo Modeling to the Next Level</i>	

In the Media

DESI Blog: At the Big Reveal: DESI’s December 2024 Unblinding Results

Podcast: Particle Mysteries: The Coldest Case - “Chasing Shadows”

YouTube: Science 101: What are dark matter and dark energy?

**Skills &
Experience**

Programming Languages: PYTHON, C, BASH, GIT, L^AT_EX

Misc.: Jax, scikit-learn, emcee, GADGET, CAMB, 2LPTIC, ROCKSTAR

Parallel Computing: MPI, OPENMP

Observing Experience:

~ 80 hours using 2.3 meter telescope at Wyoming Infrared Observatory

~ 80 hours using 0.6 meter telescope at Williams College

~ 200 hours using 0.6 meter telescope at Wellesley College

~ 100 hours using 8'' reflector telescopes at Wellesley College and Vanderbilt University

~ 100 hours using 6'' and 12'' historic refractor telescopes at Wellesley College

**Public Service
& Outreach**

Future Physicists – 5 College Women+ Career Day, Smith College

Jan. 2026

DESI Mentorship Program

Oct. 2023 -

Science Careers in Search of Women panelist, Argonne

March 2023

Conference for Undergraduate Women in Physics, Argonne

Jan. 2023

AAS Congressional Visits Day (virtual)

Sept. 2020

Science Day with Nashville Girl Scout Troop

March 2019

Meet the Astronomer Night at Dyer Observatory

Oct. 2018

Vanderbilt Student Volunteers for Science

Fall 2016

Whitin Observatory Volunteer, Wellesley College

2012–2016

Collaborations

Dark Energy Spectroscopic Instrument (DESI)

2022–

C3 and GQC Working Groups

Alternative Clustering Methods Topical Group (co-leader, November 2024-)

LSST Dark Energy Science (DESC)

2022–

Large Suite of Dark Matter Simulations (LasDamas)

2017–2022

Co-Investigator & XSEDE Allocation Manager

American Astronomical Society

2015–

**Professional
Service**

NSF Review Panel

April 2026

Scholarly Journal Peer Reviewer

Monthly Notices of the Royal Astronomical Society

Astronomy & Astrophysics

Journal of Cosmology and Astroparticle Physics

Physics of the Dark Universe